

GDP Measurement: Three Approaches Study Notes

Overview

This example demonstrates how Gross Domestic Product (GDP) can be measured using three equivalent approaches, all yielding the same result (\$50,000) despite measuring different aspects of economic activity.

Company Financial Details

OrangeInc Transactions

Transaction Type	Amount
Revenue Sources	
Oranges sold to public	\$10,000
Oranges sold to JuiceInc	\$25,000
Total Revenue	\$35,000
Expenses	
Wages paid to employees	\$15,000
Taxes paid to government	\$5,000
Total Expenses	\$20,000
Before-tax Profit	\$20,000
After-tax Profit	\$15,000

JuiceInc Transactions

Transaction Type	Amount
Revenue Sources	
Revenue from orange juice sales	\$40,000
Total Revenue	\$40,000
Expenses	
Oranges purchased from OrangeInc	\$25,000
Wages paid to employees	\$10,000
Taxes paid to government	\$2,000
Total Expenses	\$37,000
Before-tax Profit	\$5,000
After-tax Profit	\$3,000

Comparative Summary

Metric	OrangeInc	JuiceInc	Total
Total Revenue	\$35,000	\$40,000	\$75,000
Total Wages Paid	\$15,000	\$10,000	\$25,000
Total Taxes Paid	\$5,000	\$2,000	\$7,000
Before-tax Profits	\$20,000	\$5,000	\$25,000
After-tax Profits	\$15,000	\$3,000	\$18,000
Value Added	\$35,000	\$15,000	\$50,000

Approach 1: Product (Value-Added) Approach

Concept: Sum the value added at each stage of production to avoid double counting.

Value-Added Calculation Table

Company	Total Output	Purchased Inputs	Value Added
OrangeInc	\$35,000	\$0	\$35,000
JuiceInc	\$40,000	\$25,000	\$15,000
Total GDP			\$50,000

Key Point: We subtract intermediate goods to prevent counting the \$25,000 of oranges twice.

Approach 2: Income Approach

Concept: Sum all income earned by factors of production (wages + profits).

Method A - Before-tax Incomes

Income Source	OrangeInc	JuiceInc	Total
Profits	\$20,000	\$5,000	\$25,000
Wages	\$15,000	\$10,000	\$25,000
Total GDP			\$50,000

Calculation Details:

- OrangeInc profits: $\$35,000 - \$15,000 = \$20,000$
- JuiceInc profits: $\$40,000 - \$25,000 - \$10,000 = \$5,000$
- Total wages: $\$15,000 + \$10,000 = \$25,000$

Method B - After-tax Incomes + Government Revenue

Income Source	Amount
OrangeInc after-tax profits	\$15,000
JuiceInc after-tax profits	\$3,000
Total wages (no worker taxes)	\$25,000
Government tax revenue	\$7,000
Total GDP	\$50,000

Approach 3: Expenditure Approach

Concept: Sum spending by ultimate users of final goods and services.

Expenditure by Ultimate Users

Final Good	Purchased By	Amount
Fresh oranges	Households	\$10,000
Orange juice	Households	\$40,000
Total GDP		\$50,000

Key Point: Juicelnc is not an ultimate user because it transforms oranges into juice for final consumption.

GDP Calculation Summary

Approach	Method	Result
Product Approach	Sum of value added	\$50,000
Income Approach A	Before-tax wages + profits	\$50,000
Income Approach B	After-tax incomes + taxes	\$50,000
Expenditure Approach	Final goods spending	\$50,000

Important Concepts

Double Counting Problem

- **Wrong calculation:** Simply adding all output ($\$35,000 + \$40,000 = \$75,000$) would overstate economic activity
- **Problem:** This counts the \$25,000 of intermediate oranges twice
- **Solution:** Use value-added approach to count each dollar of production only once

GDP Identity

All three approaches measure the same economic activity from different perspectives:

- **Product:** What is produced (value added)
- **Income:** What is earned from production
- **Expenditure:** What is spent on final goods

Final vs. Intermediate Goods

- **Final goods:** Consumed by ultimate users (households buying oranges and juice)
- **Intermediate goods:** Used in further production (oranges sold to Juicelnc)
- **Rule:** Only final goods count in expenditure approach

Key Takeaway

This demonstrates the fundamental accounting identity in macroeconomics:

GDP = Total Value Added = Total Income = Total Expenditure on Final Goods

All three methods yield \$50,000, proving that different measurement approaches capture the same underlying economic reality.